DATA ANALYTICS MINI-PROJECT

SikaLink: Mobile Money & Micro-Lending Performance Review

Project detail	Information
Issued	Tuesday, 3 November 2026
Deadline	Friday, 6 November 2026, by 11:59pm
Format	Individual assignment. Each student submits their own work.
Tools	SQL (SQL Server, MySQL or PostgreSQL) and Power BI Desktop
Data	5 CSV files supplied (synthetic data, no real customers). Your role: junior analyst at SikaLink

1. The Scenario

SikaLink is a fast-growing Ghanaian fintech. Customers use its app and USSD code to deposit and withdraw cash through agents, send money (P2P), pay bills, pay merchants and buy airtime. Since mid-2025 it also offers short-term loans: Quick Cash, Biz Boost and Salary Advance.

It is January 2026 and the Head of Product must present the 2025 year-end review to the executives. She has heard three claims and does not know which are true:

- "Our USSD channel fails too often and we are losing customers because of it."
- "Our agent network is growing, but a small number of agents do most of the work."
- "Lending is profitable, but defaults are climbing in some regions and products."

Your job: use the data to confirm or challenge each claim. Your dashboard should make the answer clear to a busy executive.

2. How the Two Tools Work Together

STEP 1: SQL Clean the data and answer 8 questions	**STEP 2: POWER BI** Build a 2-page dashboard
Output: clean tables and 8 query results (CSV)	Output: a .pbix report whose numbers match your SQL results

3. The Data

All amounts are in Ghana cedis (GHS). Transactions cover 1 January to 31 December 2025. Loan repayment status is as at 31 December 2025.

File	Rows	What it contains
customers_raw.csv	4,045	customer_id, signup_date, region, age_group, gender, acquisition_channel, kyc_level
transactions_raw.csv	50,180	txn_id, customer_id, agent_id, txn_date, txn_type, amount_ghs, fee_ghs, channel (USSD or App), status (Success, Failed, Reversed)
agents.csv	150	agent_id, agent_region, agent_type (Kiosk, Shop, Mobile Agent)
loans.csv	2,500	loan_id, customer_id, disbursement_date, product, principal_ghs, term_days, interest_rate, status
repayments.csv	3,398	repayment_id, loan_id, due_date, paid_date, amount_due_ghs, amount_paid_ghs (empty paid_date = not yet paid)

Warning: the raw files are messy on purpose

Expect duplicate rows (customers and transactions), region names typed in different ways, dates in mixed formats, amounts stored as text, and missing customer fields. There may be others. Keep a short cleaning log: issue found, rows affected, how you fixed it.

4. Instructions and Timeline

SQL skills expected: SELECT, WHERE, GROUP BY, HAVING, ORDER BY, JOINs, CASE WHEN, date functions, and the aggregates COUNT, SUM, AVG, MIN and MAX. CTEs and window functions are not required.

Phase 1 (Tue 3 to Wed 4 Nov): SQL cleaning and analysis

- Create a database called sikalink and import the 5 CSV files into tables with a "raw_" prefix (for example raw_transactions). Never edit the raw tables.
- Check the data: count rows, count distinct IDs, and look for NULLs, blanks and odd values in each table.
- Clean into new tables called customers and transactions (for example with SELECT DISTINCT, TRIM, UPPER or LOWER, CAST). Show before and after row counts.
- Answer the 8 questions below with one query each, using successful transactions only unless stated. Save each result as q1.csv to q8.csv and put all queries in one commented .sql file.

No.	Business question
Q1	For each month of 2025: number of transactions, total value and number of distinct active customers.
Q2	Success rate and failure rate for USSD versus App (use all statuses for this one).
Q3	The top 10 agents by transaction value. Then list agents with no transactions at all (hint: LEFT JOIN).
Q4	Fee revenue by transaction type and by region.
Q5	New customers per month by acquisition channel.
Q6	Loans by product: number of loans, total lent and average loan size. Repeat by region.
Q7	By product: percentage of repayments paid on time and average days late (hint: CASE WHEN and date difference).
Q8	Default rate (status = Defaulted) by product, by region and by KYC tier. Use HAVING to show only groups with at least 30 loans.

Phase 2 (Wed 4 to Thu 5 Nov): Power BI dashboard

- **Load** the clean tables and add a Date table (DAX: CALENDAR) marked as the date table.
- **Model:** connect Customers, Agents, Loans and Date to the fact tables (Transactions, Repayments) with one-to-many relationships. Take a screenshot of the model view.
- **Write these measures** in a separate measures table: Total Value, Transaction Count, Fee Revenue, Success Rate %, Active Customers (DISTINCTCOUNT), Total Disbursed, On-Time Repayment %, and Default Rate %. Use SUM, COUNTROWS, DIVIDE and CALCULATE.
- **Page 1, Transactions and Agents:** KPI cards, monthly trend, value by transaction type, USSD versus App success rate, top 10 agents.
- **Page 2, Lending and Risk:** disbursement by product, default rate by region and product, repayment trend.
- **Add slicers** for month, region and channel. Each page title should state a finding (for example "USSD fails more than twice as often as the App").
- **Reconcile:** your Power BI totals must match your SQL results for Q1, Q2 and Q8. Note any difference and explain it.

Phase 3 (Fri 6 Nov): Final checks and submission

Re-run your queries, confirm that your Power BI numbers match your SQL results, check that every file opens, and submit before the deadline.

5. What to Submit

Zip one folder named Surname_Firstname_SikaLink and submit it by Friday 6 November, 11:59pm.

- **SQL:** one commented .sql file, the 8 result CSVs (q1 to q8) and your cleaning log
- **Power BI:** SikaLink_Dashboard.pbix, plus screenshots of both pages and the model view

6. Rules and Tips

- **This is individual work.** You may discuss ideas with friends, but your queries and dashboard must be your own. Identical files will be investigated.
- **Ask early.** If a question is unclear, ask by Wednesday, not on deadline day.
- **Think like an analyst.** A correct chart with no insight is not useful. Always ask: so what does this mean for SikaLink?

Good luck. Show me how you think, not only what you click.